Assessment Task.

Assessment Task Notification
Assessment 1 **Due Date:** Term 2 Week 7 (Thu 4 Jun '26) **Weighting:** 30%

Submission instructions:
Production portfolio and project to be submitted in class or to the TAS staffroom no later than lunch 2 on due date.
See school assessment policy for extensions and other assessment modifications.

Outcomes being assessed:

P1.2	identifies appropriate equipment, production and manufacturing techniques, including new and developing technologies
P3.2	applies research and problem-solving skills
P4.1	demonstrates a range of practical skills in the production of projects
P4.2	demonstrates competency in using relevant equipment, machinery and processes
P4.3	identifies and explains the properties and characteristics of materials/components through the production of projects
P5.1	uses communication and information processing skills
P7.1	identifies the impact of one related industry on the social and physical environment

Assessment Instructions:

Design situation:
With growing awareness of environmental sustainability and the need to reduce industrial waste, designers and fabricators are increasingly turning to reclaimed and repurposed materials to create functional and artistic works. In the metal and engineering industries, obsolete or broken machinery often contains valuable components—such as gears, springs, shafts, and housings—that can be creatively reimagined and reused.

This project challenges students to explore the artistic and functional potential of scrap metal by designing and producing an ornamental figurine that incorporates at least one significant salvaged component from old or broken machinery. The figurine should be structurally sound, visually engaging, and demonstrate competent use of a range of metalworking techniques.

Design brief:
Design and produce an ornamental metal figurine that incorporates at least one major component salvaged from broken or discarded machinery (e.g. a cog, spring, coupling, housing). The figurine should creatively highlight the salvaged part as a key feature of the design, supported by newly fabricated metal elements as needed.

You must use a minimum of two different joining methods, (e.g. welding, folding, screwing, or mechanical fastening) and may need to manufacture or modify custom parts through processes like turning, shaping, or forming bar or sheet metal.

All stages of the design and production process must be documented in a comprehensive production folio, including design sketches, material selection, joining techniques, tools and equipment used, step-by-step fabrication process, production timeline, journalling work completed and evaluation throughout production.

This project will be evaluated on two components: the final product (20%) and the production folio (80%). To achieve the highest marks, students are encouraged to consistently update their progress notes and document their reflections throughout the project.

Marking Criteria:
The following tasks in the production folio contribute to each achievement criteria:

- Recycling research (P7.1)

- Design ideas (P3.2)
    - Initial idea sketches
    - Annotations showing analysis and reflection on ideas

- Materials investigated for project (P4.3, P5.1)
    - PMI analysis of components
    - Identification of component materials
    - Mind map or evaluation of joining processes needed for component

- Production plan (P1.2)
    - List processes, tools, PPE,
    - Breakdown of production steps
    - Estimate of time

- Production journal (P4.1)
    - Daily progress notes
    - Reflection/s
    - Final evaluation of project and production processes

As well as an evaluation of your scrap metal figuring -Project (P4.2) – marked on techniques used, smooth edges, clean welds etc.

Marking Rubric:

	Outstanding (100-85%)	(85-70%)	(70-60%)	(60-50%)	Resubmit (>50%)
Recycling Research 10%	Provides a well-supported critique and appraisal of steel and metal recycling methods, including environmental and economic impacts. Defends and justifies opinions with detailed evidence about sustainability and industry practices. Demonstrates comprehensive insight into challenges and benefits of recycling metals. (8.5-10 marks)	Clearly compares and contrasts different steel and metal recycling processes, highlighting their efficiencies, limitations, and environmental outcomes. Demonstrates solid understanding of metal lifecycle and reuse strategies. (7-8.4 marks)	Classifies and describes common steel and metal recycling methods and basic environmental benefits. Shows understanding of material properties and recycling significance. (6-6.9 marks)	Defines and briefly describes steel and metal recycling concepts and processes, with basic accuracy but limited depth or insight. (5-5.9 marks)	Provides minimal, incomplete, or inaccurate information showing limited understanding of steel and metal recycling methods and their importance. (0-4.9 marks)
Design Ideas And Material Investigation 25%	Produces detailed, developed drawings that show clear idea generation and style development. Annotations provide insightful evaluation and reflection on design ideas, including what was included or excluded and why. Conducts a thorough PMI analysis of components. Demonstrates a deep understanding of component materials and provides a detailed evaluation of joining processes with	Produces multiple design sketches with annotations showing thoughtful analysis and reflection on ideas. Completes a PMI analysis and identifies component materials. Evaluates joining processes clearly, explaining suitability and reflects on the inclusion/exclusion of components in the final design. (17.5-20.9 marks)	Provides at least two initial design sketches that show idea generation. Includes basic annotations that describe ideas. Identifies component materials and joining methods, demonstrating understanding of their role in construction. Offers a simple evaluation or mind map of joining processes related to the project. (15-17.4 marks)	Provides two initial design sketches showing early ideas with limited or no annotations. Lists materials and joining processes with minimal explanation. Provides basic identification of components and joining processes without evaluation. (12.5-14.9 marks)	Sketches are missing or minimal with no development or annotations. Provides little or no information on materials or joining processes. No evidence of analysis or evaluation. (0-12.4 marks)

	Outstanding (100-85%)	(85-70%)	(70-60%)	(60-50%)	Resubmit (>50%)
	justification of their suitability for the final product. Explains what makes materials/components interesting or appropriate for the design. (21-25 marks)				
Production Plan 20%	Evaluates and justifies the selection of appropriate tools, processes, and PPE with insight. Synthesises and weighs production steps with clear reasoning. Creates a detailed and realistic timeline that reflects logical sequencing, dependencies, and workshop constraints. Demonstrates high-level planning and foresight. (17.1-20 marks)	Analyses and compares tools, processes, and PPE relevant to the task. Distinguishes and examines the steps of production with sound understanding. Constructs a realistic and mostly logical timeline, showing awareness of workflow and production stages. (14.1-17 marks)	Classifies and describes appropriate tools, processes, and PPE. Outlines and sequences production steps with general accuracy. Applies planning skills to develop a basic but workable timeline, though some elements may be unrealistic or unclear. (12.1-14 marks)	Lists or states appropriate tools, processes, and PPE relevant to the task. Identifies key production steps in a basic sequence. Includes a timeline that shows initial planning and an awareness of process stages. (10-12 marks)	Fails to recall or list key tools, processes, or PPE. Production steps are missing or irrelevant. Timeline is absent, incorrect, or unrealistic. (0-9.9 marks)
Production Journal 25%	Maintains daily progress notes that classify, organise, and compare actual work against the production plan consistently. Conducts multiple reflections, either weekly or after major project milestones, showing thoughtful insights. Provides a final evaluation that critically compares and contrasts the planned and actual	Keeps daily progress notes that contrast and classify the work done relative to the plan. Completes a midway reflection demonstrating awareness of progress and challenges. Produces a final evaluation that compares and contrasts planned versus actual outcomes and comments on production processes.	Records daily progress notes that define tasks completed without detailed comparison to the plan. Provides ongoing reflections that note general progress but lack depth. Final evaluation describes outcomes with some reference to the plan but limited critical analysis. (15-17.4 marks)	Provides notes that briefly state or list activities completed for more than half of days work occurred. Reflections are minimal or absent. Final evaluation compares and contrasts the plan and outcome but with limited explanation or evaluation. (12.5-14.9 marks)	Progress notes are missing for two-thirds or more of the days work occurred. Reflections are not recorded. Final evaluation is absent or very superficial, showing little connection to the plan or production process. (0-12.4 marks)

	Outstanding (100-85%)	(85-70%)	(70-60%)	(60-50%)	Resubmit (>50%)
	outcomes, and thoroughly evaluates production processes for strengths and areas of improvement. (21-25 marks)	(17.5-20.9 marks)			
Project 20%	Demonstrates excellent craftsmanship with clean, precise joining methods and accurately formed angles fully matching the design intent. Manufactures or modifies custom parts skillfully. Shows high creativity in the innovative, seamless incorporation of salvaged components. Any differences from the initial design are clearly recorded and justified in documentation. (17.1-20 marks)	Produces well-executed joining methods and accurate angles consistent with the design. Manufactures or modifies parts competently. Incorporates salvaged components thoughtfully and creatively. Differences from the proposed design are noted and explained. (14.1-17 marks)	Joining methods and angles are mostly clean but may show clear residue or inconsistencies such as soot, rough edges, or incomplete filing. Minor differences from the proposed design are present but not recorded in documentation. (12.1-14 marks)	Joining methods and angles show noticeable flaws; residue or roughness from processing is evident. Finished product has major differences from the proposed design, which are unrecorded. Minimal manufacturing or modification of parts. (10-12 marks)	Design is missing the minimum two required joining processes. Little to no manufacture or modification of parts. No documentation of design changes. (0-9.9 marks)

Total marks: / 100

Assessor Comments: